DO NOW · AO1 REVISION

Fleming's left-hand rule and motors

Magnetism and electromagnetism
Paper 2 · Combined and Separate
13 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 7 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**Describe the motor effect.**	A current-carrying conductor in a magnetic field experiences a force.	☐
2 ★	**In Fleming's left-hand rule, state what the thumb represents.**	The direction of the force	☐
3 ★	**In Fleming's left-hand rule, state what the first finger represents.**	The direction of the magnetic field	☐
4 ★	**In Fleming's left-hand rule, state what the second finger represents.**	The direction of the current	☐
5 ★	**Give three ways to increase the force on a current-carrying wire in a magnetic field.**	Increase the magnetic flux density, the current, or the length of wire in the field.	☐
6	**Define magnetic flux density.**	A measure of the strength of a magnetic field.	☐
7	**State the standard unit of magnetic flux density.**	Tesla (T)	☐
8 ★	**Describe what happens to a current-carrying coil in a magnetic field.**	It rotates, because the forces on its two sides act in opposite directions.	☐
9	**Name the effect used in electric motors.**	The motor effect	☐
10	**State the equation for the force on a current-carrying conductor at right angles to a magnetic field.**	force = magnetic flux density × current × length (F = B I l)	☐
11	**Give two ways to reverse the force on a current-carrying wire in a magnetic field.**	Reverse the current or reverse the magnetic field.	☐
12	**Give two ways to make an electric motor spin faster.**	Increase the current, or use a stronger magnetic field.	☐

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	QUESTION	ANSWER	✓
13 ★	**Explain why a current-carrying wire in a magnetic field experiences a force.**	The magnetic field around the wire interacts with the field it is placed in.	☐

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